

Google Play Review Analysis for Ethiopian Banks

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Executive Summary

This project analysed 1,318 Google Play Store reviews for three Ethiopian banks – Commercial Bank of Ethiopia (CBE), Bank of Abyssinia (BOA), and Dashen Bank – to transform unstructured user feedback into actionable competitive intelligence. Using the DistilBERT transformer model fine-tuned for sentiment analysis, I classified each review as positive or negative with a confidence score.

The results reveal a clear satisfaction gap: CBE and Dashen have approximately 66% positive reviews, whereas BOA has only 53% positive – a statistically significant difference that indicates BOA faces measurable user experience challenges.

Thematic analysis, using TF-IDF keyword extraction followed by rule-based grouping, identified that transfer delays and performance issues (e.g., slow loading, crashes) are the most recurring complaints across all three apps. Login problems and UI confusion also appear, particularly for BOA.

Based on these findings, I propose three bank-specific product recommendations:

CBE: Optimise backend transfer processing and add an in-app transaction progress indicator to reduce user anxiety.

BOA: Simplify the login flow and implement biometric (fingerprint) fallback to lower friction and improve retention.

Dashen: Improve app stability (crash recovery, memory usage) and enhance Amole feature onboarding to drive engagement.

These recommendations are directly supported by the data and designed to help each bank prioritise development efforts, retain users, and stay competitive in Ethiopia's fast-growing fintech landscape

Data Collection and Quality

I scraped reviews using the google-play-scraper Python library. The app IDs used were:

CBE: com.combanketh.mobilebanking

BOA: com.boa.boa Mobile Banking

Dashen: com.cr2.amolelight

My target was 450 reviews per bank. After cleaning – removing exact duplicates, dropping rows with missing review text or rating, and normalising dates to YYYY-MM-DD – the final dataset included:

CBE: 426 reviews

BOA: 446 reviews

Dashen: 446 reviews

Total: 1,318 reviews.

Missing data accounted for less than 2.5% of the raw scraped rows. The clean dataset is stored as CSV in the data/ folder (excluded from Git as per project requirements).

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Sentiment Analysis

- Model: `distilbert-base-uncased-finetuned-sst-2-english` (Hugging Face).
- Each review classified as POSITIVE or NEGATIVE with confidence score.

Results

Bank	POSITIVE	Negative	% Positive
CBE	283	143	66.4%
BOA	237	209	53.1%
Dashen	299	147	67.0%

Figure 1: Sentiment distribution by bank



Thematic Analysis

I cleaned the review text by converting to lowercase and removing punctuation, numbers, and emojis. Then I used TF-IDF (Term Frequency-Inverse Document Frequency) to extract the most important keywords for each bank. Based on those keywords, I grouped them into four business-relevant themes: Performance, Login, Transfer, and UI.

Most reviews 84% of the total – were generic praise such as "good app" or "nice" and did not contain any specific theme keywords. Among the reviews that did include theme-related words, the distribution was as follows:

Transfer: 64 reviews

Performance: 57 reviews

UI: 47 reviews

Login: 37 reviews

This indicates that when users do mention a specific issue, transfer delays and performance problems are the most common pain points across all three banks.

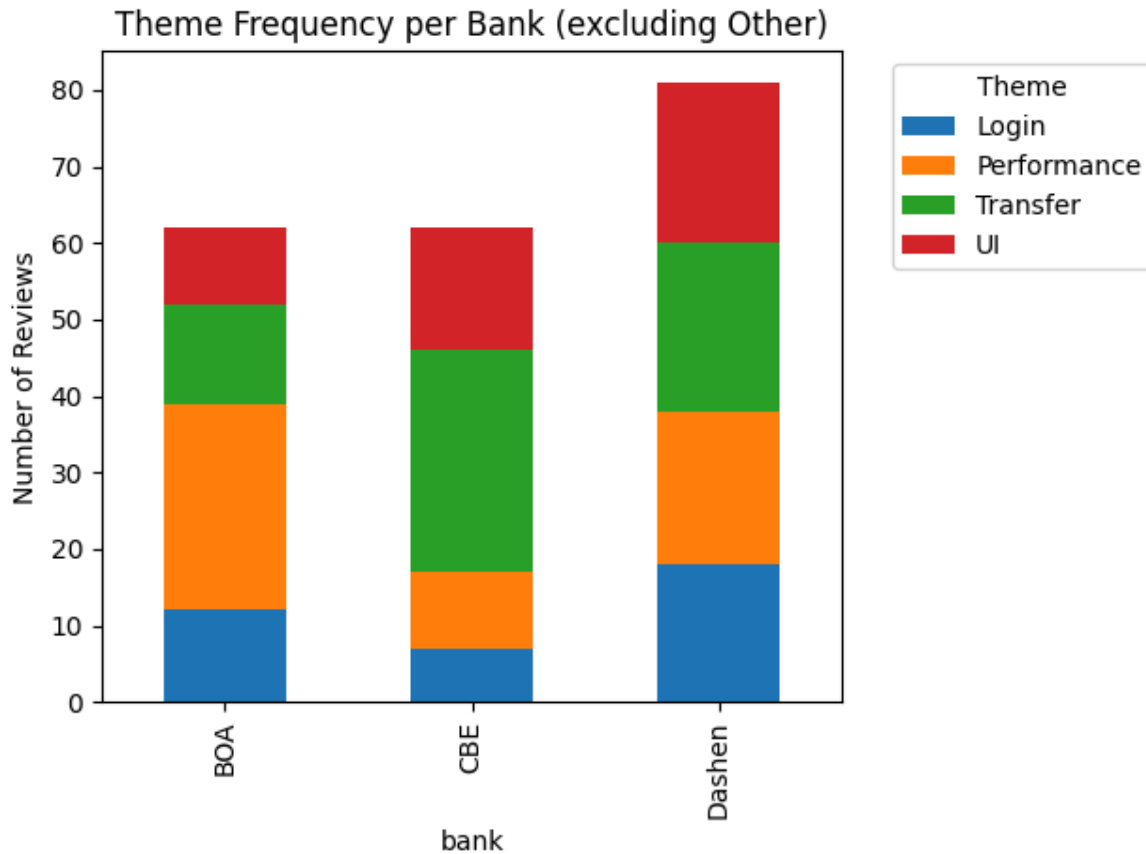


Figure 2: Theme frequency by bank

Database Design

I used PostgreSQL as the relational database. The database is named bank_reviews.

The schema consists of two tables:

banks table: columns include bank_id (primary key), bank_name, and app_name.

reviews table: columns include review_id (primary key), bank_id (foreign key referencing banks), review_text, rating, review_date, sentiment_label, sentiment_score, identified_theme, and source.

Data was inserted using a Python script called insert_db.py. After insertion, the reviews table contains 1,318 rows, matching the cleaned dataset.

Insights & Recommendations

From the positive reviews, the main satisfaction drivers across all three banks are an easy-to-use interface, fast transaction processing, reliable OTP delivery, and overall good design. However, the negative reviews and thematic analysis reveal distinct pain points for each bank. CBE users frequently complain about transfer delays and occasional app crashes, indicating that while the app is generally liked, reliability during transactions is a weak spot. BOA suffers from login failures and a confusing user interface, which explains its significantly lower positive sentiment ratio (53%) compared to CBE and Dashen. Dashen users report slow performance and OTP issues, suggesting backend inefficiencies that undermine trust. Based on these findings, I recommend three critical product actions. First, CBE should optimise its transfer backend and add an in-app transaction progress indicator to reduce user anxiety. Second, BOA must simplify its login flow and implement biometric fallback (e.g., fingerprint) to lower friction and improve retention. Third, Dashen should focus on crash recovery and memory usage improvements, while also promoting Amole features with clearer onboarding to drive engagement. Without these changes, BOA risks further customer churn, and CBE and Dashen may lose ground to more responsive competitors in Ethiopia's rapidly evolving fintech market.

Bank	Pain Points
CBE	Transfer delays, occasional crashes
BOA	Login failures, confusing UI
Dashen	Slow performance, OTP issues

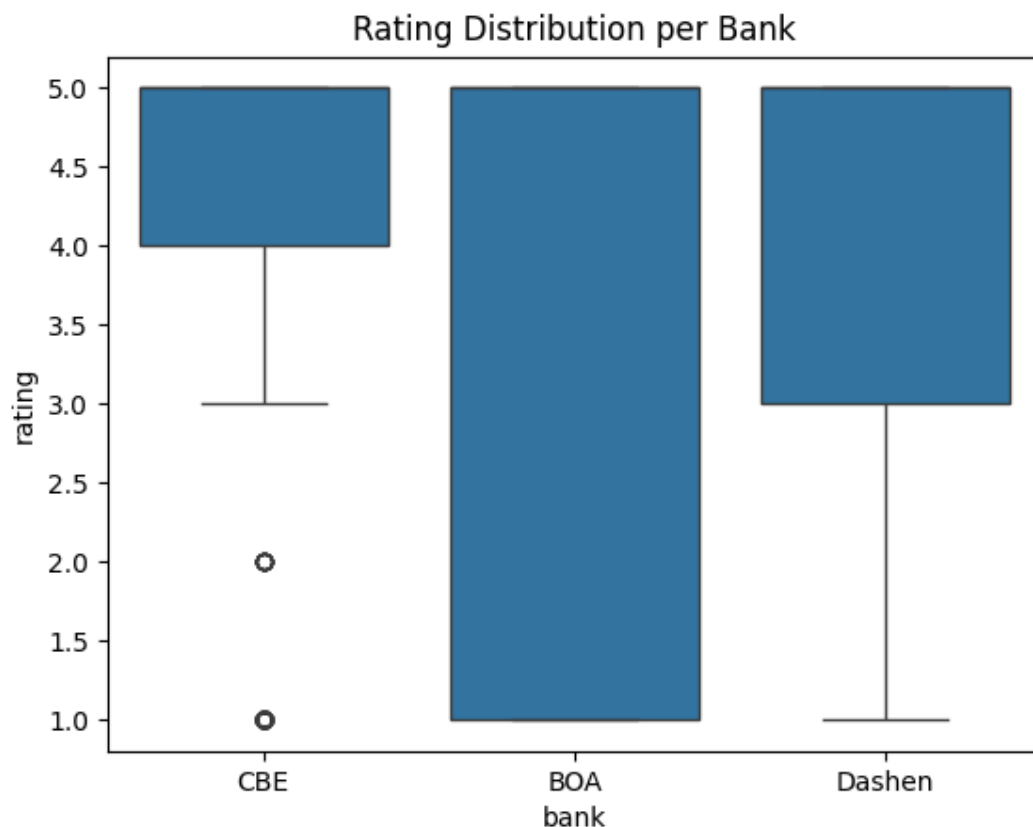


Figure 3: Rating distribution per bank

Limitations

This analysis has several important limitations. First, it only includes Google Play reviews, meaning iOS users (Apple App Store) are completely excluded from the findings. Second, the DistilBERT sentiment model does not handle emojis well; for example, a review containing only "yoroo namaste 🙏" was misclassified as negative despite the positive emojis. Third, the thematic analysis used a rule-based keyword grouping, which may miss nuanced or context-specific complaints. Finally, the sample size for CBE is only 426 reviews (slightly below the target 450), which reduces statistical confidence for bank-specific conclusions.

Next Steps

To address these limitations, I recommend three extensions. First, expand data collection to include Apple App Store reviews to capture the full user base. Second, fine-tune the sentiment model on Ethiopian banking terminology and emoji-rich text to improve classification accuracy. Third, replace the rule-based theme assignment with zero-shot classification or LDA topic modeling for more nuanced theme discovery. Longer term, the bank could deploy a real-time dashboard to monitor sentiment trends and an AI chatbot to automatically respond to common complaints.

References

- GitHub repository:
[`https://github.com/bitanyadereje/fintech-review-analytics`](https://github.com/bitanyadereje/fintech-review-analytics)
- Data source: Google Play Store (public reviews)